TARGETED APPLICATION PORTFOLIO

AMMAR AHMED

Mechanical Design & Prototyping Engineer

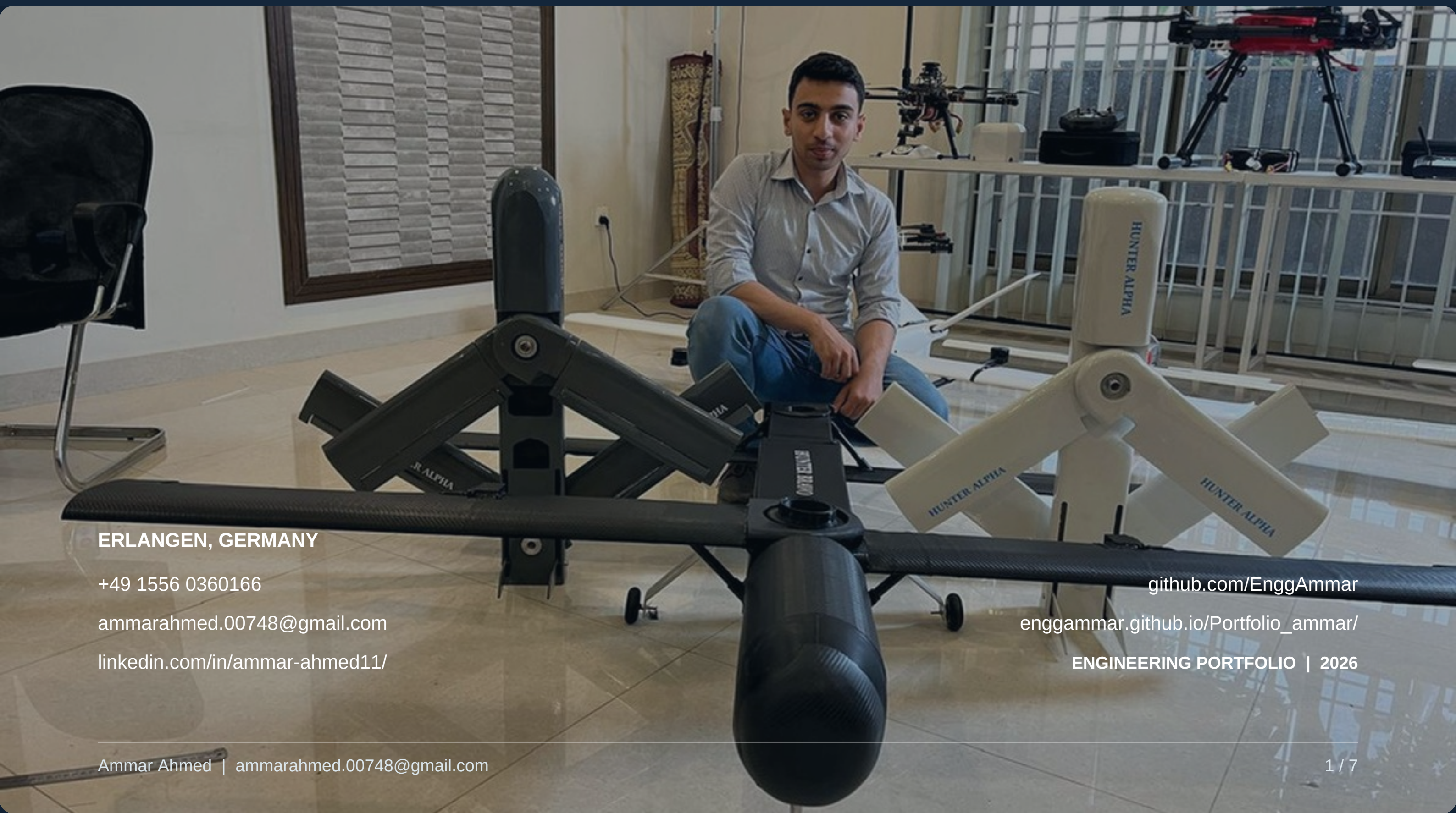
UAV Systems | Hardware Integration | Rapid Prototyping

PROTOTYPING TECHNICIAN (M/F/D)

Quantum-Systems GmbH | Gilching, Bayern

PROFILE

Mechanical engineer with hands-on experience in UAV prototype assembly, FDM 3D printing, mechanical-electronic integration, CAD-driven design, functional validation, and rapid prototype iteration. Experienced with fixed-wing UAV structures, drone demonstration models, sensors, actuators, propellers, landing gear, and practical prototype troubleshooting.



ERLANGEN, GERMANY

+49 1556 0360166

ammarahmed.00748@gmail.com

linkedin.com/in/ammar-ahmed11/

github.com/EnggAmmar

enggammar.github.io/Portfolio_ammar/

ENGINEERING PORTFOLIO | 2026

ROLE ALIGNMENT

Why I Fit Quantum-Systems

Evidence-led alignment to hands-on prototype assembly, manufacturing, hardware integration, and functional validation for unmanned aircraft systems.

Drone prototype assembly

Built fixed-wing and demonstration UAV structures.

UAV structures & airframes

Foldable structures, wings, landing gear, assembly.

Electronics & actuators

Integrated electronics, sensors, propellers, and actuators.

FDM rapid prototyping

PLA/ABS printing, printer operation, part preparation.

CAD-to-prototype

Moved concepts and CAD files into physical hardware.

Testing & troubleshooting

Fit checks, system tests, validation, issue refinement.

Design iteration

Updated geometry and assemblies from build feedback.

Composite support

Supported glass-fiber UAV structure direction.

Cross-functional work

Linked mechanical, electronics, analysis, and review.

TOOLS AND WORKING METHODS

SolidWorks · Siemens NX · CATIA v6 · ANSYS Mechanical · ANSYS Fluent · MATLAB · Python · FDM 3D Printers · Hand Tools

RELEVANT HANDS-ON EXPERIENCE

NASTP | Research Officer	Prototype assembly at scale; demonstration drone integration; composite structure support.
Aero-Vision | UAV Prototyping	Foldable fixed-wing UAV design; FDM manufacturing; electronics/sensor integration; CFD support.
WS Audiology | Commissioning	Structured subsystem and system testing; documentation; repeatable validation workflows.

CASE STUDY 01

UAV Design, Fabrication & Integration

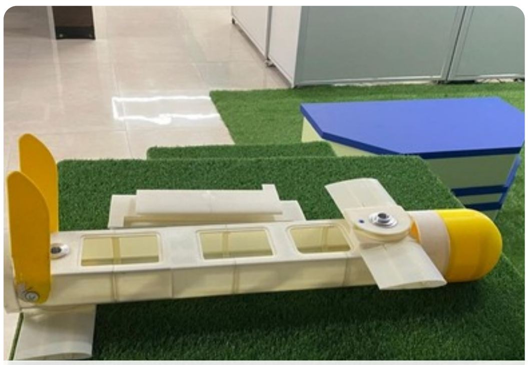
Assembled fixed-wing foldable UAV airframes prepared as demonstration-ready engineering prototypes.

Fixed-wing foldable UAV prototyping covering CAD-driven structure development, FDM prototype manufacturing, electronics integration, aerodynamic simulation support, and system-level assembly for flight-ready prototypes.

- Built and assembled UAV prototype structures from engineering concepts and CAD design files.
- Integrated mechanical assemblies with electronics, propellers, actuators, landing gear, and sensor-related hardware.
- Produced FDM 3D printed UAV components and supported fit checks before final assembly.
- Supported aerodynamic validation through CFD/mesh preparation and prototype refinement.
- Prepared demonstration-ready UAV models for engineering review and stakeholder presentation.

UAV Prototyping | Drone Assembly | FDM 3D Printing | Hardware Integration | CAD | CFD | Functional Validation

Meshed UAV geometry for aerodynamic analysis.

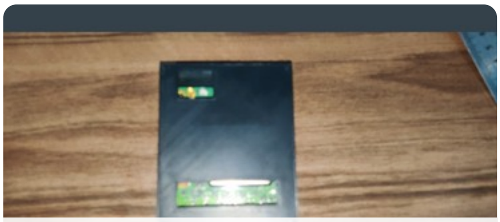


FDM UAV structure before final system assembly.

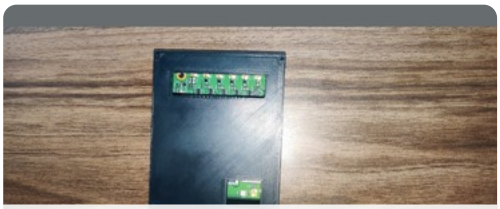
CASE STUDIES 02 + 05

Rapid Prototype Manufacturing

FDM-manufactured UAV body and mechanism used for assembly development.



Prototype printer setup.



Printer operation and build setup.

3D PRINTED UAV COMPONENTS

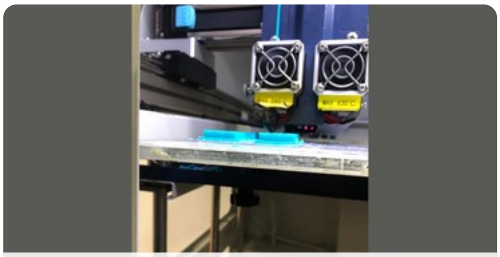
Rapid prototype manufacturing using FDM 3D printing, CAD-to-part preparation, printer operation, fit validation, and iterative design improvements for engineering prototype parts.

- Converted CAD models into printable prototype components for fast design validation.
- Operated and maintained FDM 3D printers for PLA/ABS prototype manufacturing.
- Performed fit checks, assembly refinement, and physical validation of printed parts.
- Supported rapid design iterations by moving from CAD to usable prototype hardware.
- Applied practical fabrication thinking to improve assembly quality and part usability.

3D Printing | **Rapid Prototyping** | **CAD-to-Part** | **PLA/ABS** | **Fit Checks**

MECHANICAL INTERFACE DESIGN

CAD-driven interface development, physical part validation, alignment checks, and geometry refinement based on practical assembly feedback.



Printed model for design review and fit validation.

- Designed and manufactured prototype interface parts.
- Checked fit, alignment, and usability in assemblies.
- Refined geometry from hands-on build feedback.

Mechanical Interfaces | **Assembly** | **Fit Validation**

CASE STUDIES 03 + 06

Hardware Integration & Prototype Testing

Full demonstration drone models combining structures, propulsion-related hardware, and landing systems.

Mechanism and airframe integration reference.

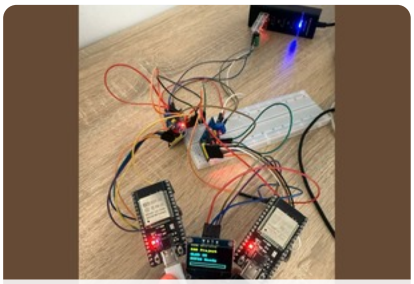
UAV ELECTRONICS, MOTORS, SENSORS & SYSTEM INTEGRATION

Hands-on UAV integration covering mechanical assembly, electronics installation, actuator and propeller integration, landing gear installation, and preparation of demonstration drone models.

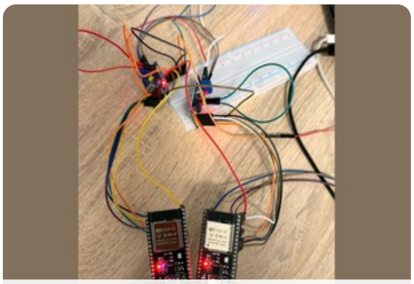
- Integrated electronics, propellers, actuators, landing gear, and structural components into full demonstration drone models.
- Supported system-level build preparation for prototype testing and review.
- Worked across mechanical and electronic interfaces to support reliable UAV assembly.
- Gained hands-on exposure to ArduPilot-style flight-control workflows where applicable.
- Helped troubleshoot integration issues during prototype development and assembly.

System Integration | Motors | Actuators | Sensors | Avionics | Prototype Testing | Troubleshooting

ELECTRONICS TESTBENCH HIGHLIGHT: HIL-INSPIRED CAN VALIDATION



ESP32/MCP2515 hardware setup.



Wired CAN testbench.

Electronics testbench using ESP32 and MCP2515 CAN hardware, wiring, signal validation, fault injection, and automated PASS/FAIL checks.

- Built and wired repeatable bench hardware.
- Validated signals with structured test cases.
- Used PASS/FAIL outputs for troubleshooting.

CASE STUDY 04

Composite UAV Structure Support

UAV structural component manufactured for prototype evaluation and next-stage build learning.

GLASS-FIBER STRUCTURE DIRECTION

Supported the transition toward glass-fiber reinforced composite UAV structures to improve prototype robustness for the next development phase.

- Supported composite structure direction for UAV prototype development.
- Helped evaluate robustness improvements for next-stage prototype builds.
- Connected manufacturing feedback with structural design requirements.
- Highlighted practical awareness of composite materials relevant to drone production.

Analysis-ready airframe geometry.

Composites | Glass Fiber | UAV Structures | Prototype Robustness | Manufacturing Support

BUILD-TO-VALIDATION LOOP

1 Concept & CAD → 2 Prototype Build → 3 Fit & Integration → 4 Functional Review → 5 Design Refinement

Prototype structures assembled for review.

APPLICATION POSITIONING

READY TO BUILD, TEST AND IMPROVE UAV HARDWARE

I am targeting hands-on prototyping, UAV assembly, hardware integration, rapid prototype manufacturing, and functional validation roles where practical build quality and engineering iteration are critical.

CORE CONTRIBUTION

- Hands-on assembly of prototype UAV structures and demonstration systems.
- Practical CAD-to-part manufacturing through FDM 3D printing and fit validation.
- Mechanical-electronic integration across sensors, actuators, propulsion-related hardware, and structures.
- Structured testing, troubleshooting, documentation, and iterative prototype refinement.

Ammar Ahmed

MECHANICAL DESIGN & PROTOTYPING ENGINEER

Erlangen, Germany | +49 1556 0360166

ammarahmed.00748@gmail.com

linkedin.com/in/ammar-ahmed11/

github.com/EnggAmmar | enggammar.github.io/Portfolio_ammar/